

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ILLUMINA INC.,
Petitioner,

v.

MOLECULAR LOOP INNOVATIONS LLC,¹
Patent Owner.

IPR2024-00965
Patent 11,768,200 B2

Before ANNETTE R. REIMERS, SUSAN L. C. MITCHELL, and
ZHENYU YANG, *Administrative Patent Judges*.

PER CURIAM.

JUDGMENT

Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

¹ After institution, Patent Owner changed its name from Molecular Loop Biosciences, Inc. to Molecular Loop Innovations LLC. *See* Paper 11, 1. We modify the caption accordingly.

I. INTRODUCTION

Illumina, Inc. (“Petitioner”) filed a Petition (Paper 1, “Pet.”) seeking *inter partes* review of claims 1–6 of U.S. Patent No. 11,768,200 B2 (Ex. 1001, “the ’200 patent”). We instituted trial to review the challenged claims. Paper 7 (“DI”).

Thereafter, Molecular Loop Innovations LLC (“Patent Owner”) filed a Response to the Petition (Paper 18, “PO Resp.”), Petitioner filed a Reply to Patent Owner’s Response (Paper 25, “Reply”), and Patent Owner filed a Sur-reply to Petitioner’s Reply (Paper 33, “Sur-reply”).

An oral hearing for this proceeding was held on September 23, 2025, and the transcript of that hearing is of record. *See* Paper 39 (“Tr.”).

The Board has jurisdiction under 35 U.S.C. § 6 and issues this Final Written Decision pursuant to 35 U.S.C. § 318 and 37 C.F.R. § 42.73. For the reasons provided below, we find Petitioner has shown, by a preponderance of the evidence, the unpatentability of claims 1–6.

A. *Related Matters*

According to the parties, the ’200 patent is involved in the following cases: *Molecular Loop Biosciences, Inc. v. Illumina, Inc.*, No. 1:24-cv-00680 (D. Del.) and *Illumina, Inc. et al. v. Molecular Loop Biosciences, Inc.*, No. 1:24-cv-11501 (D. Mass.). Pet. 70; Paper 3, 1. Patent Owner later informed us that this case before the District Court of Massachusetts is no longer pending. Paper 11, 1.

Petitioner filed IPR2024-00964, challenging the claims of U.S. Patent No. 11,041,852, a patent in the same family as the ’200 patent. Pet. 70.

B. The '200 Patent

The '200 patent relates generally to “methods for maintaining the integrity and identification of a nucleic acid template in a multiplex sequencing reaction.” Ex. 1001, Abstract. The methods include the steps of “obtaining a template nucleic acid, incorporating a pair of sequence identifiers into the template, and sequencing the template.” *Id.*

The '200 patent states that “[n]ext generation sequencing [“NGS”] typically involves the generation of an in vitro library; arraying of nucleic acid templates into physically distinct locations on a solid support (and optional clonal amplification of the templates into spatially localized clusters); and the sequencing-by-synthesis reaction itself.” *Id.* at 1:43–48.

According to the '200 patent, maintaining accurate sample identification during a multiplex sequencing reaction is an issue because “the sample preparation phase frequently introduces errors into the subsequent sequencing reaction.” *Id.* at 1:37–39. “For example, template molecules may physically overlap on a substrate; and in some cases result in a detectable erroneous signal.” *Id.* at 1:39–41; *see also id.* at 1:52–54 (stating that errors can be introduced from “improper pairing of template strands from different samples,” which can “form during the amplification of the multiplexed samples”). Because these errors have a significant effect on the results, the '200 patent states that “methods for reducing or eliminating errors in such procedures are needed.” *Id.* at 1:60–62.

The '200 patent explains the undesired introduction of sequencing errors when using a single barcode identifier. *See id.* at 4:17–63, Fig. 1 (showing “one example of how molecular overlap can lead to apparent high quality, but incorrect sequence data”). The '200 patent describes a method

that uses “two or more identifiers that are uniquely associated with an analyte of interest, such that the identifiers will only be present when the analyte of interest is detected.” *Id.* at 2:5–8. According to the ’200 patent, “[i]n this way, false positive and/or false negative results are avoided by requiring concomitant detection of two or more independent markers associated with the analyte to be detected.” *Id.* at 2:8–11.

The ’200 patent states that, using two or more nucleic acid barcodes as identifiers, its method is especially useful in multiplex NGS. *Id.* at 2:12–18. The ’200 patent explains that

in sequencing applications, a pair of barcodes are placed such that they flank the sequence of interest. The identifiers are unique to the sample, so that valid sequence data are confirmed by the presence of the pair of flanking identifiers. In contrast, template containing only one identifier or an incorrect pair (i.e., a pair or more of identifiers not associated with the same template) will be excluded from analysis. Thus, methods of the invention make it possible to determine whether the integrity of a sample was maintained during a multiplex sequencing reaction, and prevent the assigning of sequence data to the wrong sample.

Id. at 2:18–29; *see also id.* at Fig. 2, 4:64–6:6 (describing Figure 2 as “depicting how methods of the invention allow one to detect, during analysis, whether the integrity of a patient’s sample was maintained during a multiplex sequencing reaction”).

The ’200 patent states that its invention “is especially useful in multiplex next-generation sequencing applications in which errors can have a significant impact on results.” *Id.* at 2:12–14.

C. Illustrative Claim

Of the challenged claims, only claim 1 is independent. Claim 1 is illustrative of the claimed subject matter and is reproduced below.

1. A method for validating the sequence of a nucleic acid analyte of interest in a multiplex sequencing reaction, the method comprising:

i) detecting the presence of two or more identifier sequences that are uniquely associated with the nucleic acid analyte of interest,

wherein at least a first identifier sequence is incorporated into a 5' portion of the nucleic acid analyte of interest and wherein at least a second identifier sequence is incorporated into a 3' portion of the nucleic acid analyte of interest, and

wherein the first and second identifier sequences have four or more nucleotides and are different;

ii) sequencing the first identifier sequence, the nucleic acid analyte of interest and the second identifier sequence; and

iii) validating the sequence of the nucleic acid analyte of interest by analyzing both identifiers and excluding the sequences of those nucleic acid analytes of interest containing only one identifier or an incorrect pair of identifiers from sequence analysis of the multiplex sequencing reaction.

Ex. 1001, 21:2–22:6.

D. Instituted Challenges to Patentability

We instituted trial to determine whether the challenged claims are unpatentable based on the following bases:

Claims Challenged	35 U.S.C. § ²	References
1–6	102	Gloor ³
1–6	102	Parameswaran ⁴
1–6	103	Parameswaran, Gloor
1–6	103	Parameswaran, Binladen ⁵
6	103	Parameswaran, Binladen, Bentley ⁶

In support of their respective positions, Petitioner relies on the Declarations of Gregory B. Gloor, Ph.D. (Exs. 1002, 1046) and Paul T. Spellman, Ph.D. (Exs. 1003, 1054); and Patent Owner relies on the Declaration of Viktor Adalsteinsson, Ph.D. (Ex. 2001).

² The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. §§ 102 and 103, effective March 16, 2013. On its face, the ’200 patent claims priority to a provisional application filed on December 23, 2010. Ex. 1001, code 60. For purposes of this proceeding, the parties apply this date as the effective priority date. *See* Pet. 17; PO Resp. 8. We do the same. Accordingly, we apply the pre-AIA version of §§ 102 and 103.

³ G.B. Gloor, et al., *Microbiome profiling by Illumina sequencing of combinatorial sequence-tagged PCR products*, Cornell University Library, arXiv:1007.5075v1 (2010) (Ex. 1005, “Gloor”).

⁴ P. Parameswaran, et al., *A pyrosequencing-tailored nucleotide barcode design unveils opportunities for large-scale sample multiplexing*, 35 NUCLEIC ACIDS RSCH, 19(e130) (2007) (Ex. 1004, “Parameswaran”).

⁵ J. Binladen, et al., *The use of coded PCR primers enables high-throughput sequencing of multiple homolog amplification products by 454 parallel sequencing*, 2 PLoS ONE 2:e197 (2007) (Ex. 1029, “Binladen”).

⁶ D.R. Bentley, et al., *Accurate whole human genome sequencing using reversible terminator chemistry*, 456 NATURE 6, 53–59 (2008) (Ex. 1006, “Bentley”).

II. ANALYSIS

A. *Principles of Law*

To prevail in this *inter partes* review, Petitioner “shall have the burden of proving a proposition of unpatentability by a preponderance of the evidence.” 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

Anticipation is a question of fact. *In re Gleave*, 560 F.3d 1331, 1334–35 (Fed. Cir. 2009). Whether a reference anticipates a claim is assessed from the perspective of an ordinarily skilled artisan. *See Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358, 1368 (Fed. Cir. 2003). To show anticipation, each and every claim element, arranged as in the claim, must be disclosed in a single piece of prior art. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). For a claim directed to a genus, if a prior art reference discloses a species falling within the claimed genus, the species anticipates the genus. *In re Slayter*, 276 F.2d 408, 411 (CCPA 1960).

A claim is unpatentable for obviousness if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” 35 U.S.C. § 103 (pre-AIA); *see also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) when in evidence, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

An invention is not obvious just because all of the elements that comprise the invention were known in the prior art. *KSR*, 550 U.S. at 418 (“[A] patent composed of several elements is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.”). Rather, “it can be important to identify a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements in the way the claimed new invention does.” *Id.* Thus, a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (citations omitted).

In an obviousness determination, the presence or absence of a motivation to combine references, the presence or absence of a reasonable expectation of success, as well as what a reference teaches and whether it teaches toward or away from the claimed invention, are all questions of fact. *Par Pharm., Inc. v. TWI Pharm., Inc.*, 773 F.3d 1186, 1196–97 (Fed. Cir. 2014).

We analyze the instituted grounds of unpatentability in accordance with these principles.

B. Level of Ordinary Skill in the Art

In determining the level of ordinary skill in the art, various factors may be considered, including the “type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active

workers in the field.” *In re GPAC, Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995). Furthermore, the prior art itself can reflect the appropriate level of ordinary skill in the art. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

Petitioner asserts that a person of skill in the art (“POSA”) “would have had (i) a Ph.D. in molecular biology, genetics, bioinformatics, or a related field and at least two years of experience in high-throughput sequencing technologies or (ii) a Master’s degree in one of the same fields with at least four years of experience in high-throughput sequencing technologies.” Pet. 18 (citing Ex. 1003 ¶¶ 19–20).

Patent Owner states that it adopts Petitioner’s proposal for a POSA for purposes of this proceeding. PO Resp. 13 (citing Ex. 2001 ¶¶ 20–23).

After reviewing the record, we adopt Petitioner’s proposed definition of the skill level as it is consistent with the prior art’s demonstration of the level of ordinary skill in the art at the time of the invention. We find both Dr. Spellman and Dr. Adalsteinsson are qualified to opine from the perspective of a POSA, because each has at least ordinary skill in the art. *See* Ex. 1003 ¶¶ 4–15; Ex. 2001 ¶¶ 4–9.

C. Claim Construction

In an *inter partes* review, we construe a claim term “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b).” 37 C.F.R. § 42.100(b) (2020). Under that standard, the words of a claim “are generally given their ordinary and customary meaning,” which is “the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention, i.e., as of the effective filing date of the patent application.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Claim 1 recites “detecting the presence of two or more identifier sequences that are uniquely associated with the nucleic acid analyte of interest.” The parties disagree over the construction of this limitation. Reply 2–3; Sur-reply 1–6. They emphasize different aspects of the limitation: for Petitioner, “***two or more*** identifier sequences that ***are uniquely associated*** with the nucleic acid analyte of interest;” whereas for Patent Owner, “***identifier sequences []*** are ***uniquely associated*** with the nucleic acid analyte of interest.” Reply 2; Sur-reply 3.

We acknowledge the ambiguous nature of the disputed claim language. On its face, it could require two or more identifier sequences ***together*** be uniquely associated with the nucleic acid of interest, as Petitioner argues; or it could require two or more identifier sequences, each individual identifier sequence must be uniquely associated with the nucleic acid of interest, as Patent Owner asserts. *See* Reply 2–3; Sur-reply 1–6. We, therefore, must examine the words of the claim and the specification to ascertain the scope of this limitation.

The specification states that “[w]hile the pair of barcodes associated with any given template is unique, ***individual barcode sequences can be used with multiple templates*** as long as no pair is used across two different samples.” Ex. 1001, 2:50–53 (emphasis added). Elsewhere, it states that

[t]he first and second identifiers are designed such that ***each set of identifiers is correlated to a particular sample*** (e.g., material obtained from a particular patient or group of patients), allowing samples to be distinguished and/or validated. The first and second identifiers may be the same or different, ***so long as the combination is unique*** with respect to the sample or sub-sample that one seeks to distinguish and/or validate.

Id. at 3:15–22 (emphases added). We agree with Petitioner that the specification supports its proposed construction. *See* Reply 2.

Patent Owner argues that “Petitioner’s justifications for its claim construction are not compelling.” Sur-reply 3. According to Patent Owner, the language of the specification Petitioner relies on does not reflect the claimed configuration, which requires the identifier sequences be uniquely associated with the nucleic acid of interest. *Id.* Essentially, Patent Owner is arguing that the specification that does not support its claim construction should not be considered. We reject such circulation reasoning.

Patent Owner contends that claim 1 requires that “incorrect pairs of identifier sequences also be excluded.” Sur-reply 4. According to Patent Owner, “[u]nder Petitioner’s construction, the claimed method would fail to validate sequences because the ‘validated’ samples would include incorrectly identified samples that the claimed methods are aimed to eliminate.” *Id.* at 2; *see also* PO Resp. 14 (“One would not be able to discern whether pairs of identifiers were ‘incorrect’ if identifier sequences were used across different nucleic acids of interest.”). We are not persuaded by this argument.

In claim 1, the sequence of the nucleic acid of interest is validated by “analyzing both identifiers and excluding the sequences of those nucleic acid analytes of interest containing only one identifier *or* an incorrect pair of identifiers from sequence analysis of the multiplex sequencing reaction.” In its common usage, the term “or” designates alternatives. Patent Owner has not pointed to persuasive evidence to show that it acted as its own lexicographer and clearly defined the term to embrace “and.” *See Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996) (“[A] patentee may choose to be his own lexicographer and use terms in a manner other than their ordinary meaning, as long as the special definition of the

term is clearly stated in the patent specification or file history.”). Thus, under claim 1, validating the sequence of the nucleic acid of interest encompasses excluding sequences containing either only one identifier, or an incorrect pair of identifiers.

Patent Owner relies on *Sierra Wireless, ULC v. Sisvel, S.P.A.*, 130 F. 4th 1019 (Fed. Cir. 2025) for the proposition that “two limitations are not mutually exclusive where ‘the premise describing the claim language’ does not support such a conclusion.” Sur-reply 4. We are not persuaded by this argument.

The claim-at-issue in *Sierra* is directed to a method of performing an automatic repeat request in a wireless communication system, comprising, among other steps,

[c] stopping the timer when the at least one data block is received from the transmitter while the timer is running, in order to prevent a triggering of a status report before the timer expires; *and*

[d] transmitting the status report to the transmitter after the timer expires, wherein the status report comprises a positive acknowledgement indicating receipt of at least one received data block.

Id. at 1021–22 (emphasis added).

In an *inter partes* review proceeding, the Board held limitations 1[c] and 1[d] are mutually exclusive. *Id.* at 1022. The Federal Circuit determined “[t]hat conclusion does not follow from the premise describing the claim language.” *Id.* Pointing to the claim language “stopping the timer ... *and* transmitting the status report after the timer expires,” the court explained that “the plain and unambiguous language of claim 1 requires that a method, to come within the claim, must perform both limitations 1[c] and 1[d] where their preconditions apply.” *Id.*

The fact in our case is diametrically opposed to that of *Sierra*. Here, the plain and unambiguous language of claim 1 “only one identifier *or* an incorrect pair of identifiers” dictates that the validating step encompasses excluding *either* type of sequences.

In sum, in view of the claim language and the specification, we agree with Petitioner that “two or more identifier sequences that are uniquely associated with the nucleic acid analyte of interest” means two or more sequences together are uniquely associated with the nucleic acid analyte of interest, and does not require each identifier sequence be uniquely associated with the nucleic acid analyte of interest. *See* Reply 2.

Claim 5 recites “homopolymer region.” The ’200 patent specification provides an express definition for the term, stating that “the barcode sequences can be designed to have minimal or no homopolymer regions, i.e., 2 or more of the same base in a row such as AA or CCC, within the barcode sequence.” Ex. 1001, 9:6–10. Thus, we agree with Patent Owner that the term “homopolymer region” should be construed as “two or more of the same base in a row.” *See* PO Resp. 13 (citing Ex. 2001 ¶ 24).

Claim terms need only be construed to the extent necessary to resolve the controversy. *Wellman, Inc. v. Eastman Chem. Co.*, 642 F.3d 1355, 1361 (Fed. Cir. 2011). On this record and for purposes of this Decision, we see no need to expressly construe any other claim term.

D. Alleged Anticipation by Gloor

Petitioner asserts that claims 1–6 are anticipated by Gloor. Pet. 25–34. After reviewing the entire record developed at trial, and as explained below, we determine Petitioner has shown by a preponderance of the evidence that Gloor anticipates claims 1–6.

1. Gloor's Prior Art Status

Relying on the Declaration of Dr. Gloor, the first named author, Petitioner contends that Gloor was published on July 29, 2010, and thus, “is prior art under at least pre-AIA 35 U.S.C. § 102(a).” Pet. 11 (citing Ex. 1002). Dr. Gloor testifies that he submitted Gloor to arXiv.org on July 28, 2010, and received an email from arXiv.org confirming the public availability of Gloor as of July 29, 2010. Ex. 1002 ¶¶ 16, 17 (citing Ex. 1002, Exhibit E). According to Dr. Gloor, “[r]ecently, arXiv Technical Support confirmed in an email to [him] that *Gloor*’s release ‘would have been announced on Thursday, July 29, 2010.’” *Id.* ¶ 17 (citing Ex. 1002, Exhibit F). Dr. Gloor also testifies that he presented a poster at a conference held between August 31 and September 2, 2010, and the poster “provides a complete citation to” the Gloor reference. *Id.* ¶ 18 (citing Ex. 1002, Exhibit H).

In its Preliminary Response, Patent Owner argued that Petitioner has failed to show a reasonable likelihood that Gloor was publicly accessible as of December 23, 2010, the alleged priority date of the ’852 patent. Paper 6, 17–21. In our decision to institute, we found that, for purposes of institution, Petitioner has met its burden to show a reasonable likelihood that Gloor was publicly accessible before the priority date of the ’200 patent. DI 10. We, however, cautioned Petitioner that to prevail in this *inter partes* review, it must meet the preponderance of the evidence standard. *Id.* (citing 35 U.S.C. § 316(e)).

After institution, Petitioner filed a Motion for Authorization to Compel Testimony and/or Documents. Paper 12. In the motion, Petitioner explained that, before even filing the Petition, it sought, albeit

unsuccessfully, an affidavit from arXiv staff regarding Gloor's publication date. *Id.* at 1 (citing Ex. 1039). After institution, Petitioner again sought such an affidavit. *Id.* at 2 (citing Ex. 1042). According to Petitioner, another arXiv employee confirmed Gloor's public availability as of July 29, 2010, "but reiterated that arXiv would not provide an affidavit." *Id.* (citing Ex. 1043). Petitioner sought leave "to apply for a subpoena compelling testimony and documents from arXiv about Gloor's public availability and arXiv's regular practices of making references publicly available." *Id.* at 3.

We denied that motion, explaining that the requested information may be available through other means, such as "testimony regarding the internet archive service known as the Wayback Machine (<https://web.archive.org/>)" or the email arXiv.org sent to subscribers on July 29, 2010, announcing the publication of Gloor. Paper 16, 5–6.

During trial, Patent Owner continues to argue that "[b]ecause Petitioner's evidence fails to show that Gloor was publicly accessible, Petitioner fails to show that Gloor qualifies as prior art printed publication." PO Resp. 17. After considering all the evidence and arguments presented by both parties on this issue, and as explained below, we find Petitioner has shown, by a preponderance of the evidence, that Gloor qualifies as prior art.

A reference is considered publicly accessible if it is "made available to the extent that persons interested and ordinarily skilled in the subject matter or art, exercising reasonable diligence, can locate it." *Jazz Pharms., Inc. v. Amneal Pharms., LLC*, 895 F.3d 1347, 1355 (Fed. Cir. 2018). To "determine whether an interested researcher would have been sufficiently capable of finding the reference and examining its contents," we "must consider all of the facts and circumstances surrounding the disclosure."

In re Lister, 583 F.3d 1307, 1312 (Fed. Cir. 2009); *see also In re Klopfenstein*, 380 F.3d 1345, 1350 (Fed. Cir. 2004) (explaining that a determination of whether a reference qualifies as a printed publication “involves a case-by-case inquiry into the facts and circumstances surrounding the reference’s disclosure to members of the public”).

Patent Owner argues that

Petitioner fails to show that Gloor was publicly accessible before December 23, 2010 for three reasons. **First**, the scant evidence Petitioner relies upon to show public accessibility—a declaration and exhibits submitted by Dr. Gloor (EX1002), one of the authors of the Gloor reference—shows at best that Gloor may have been **submitted** to arXiv.org by July 28, 2010, but does not show that Gloor was actually disseminated or accessible to the public on July 29, 2010. **Second**, Petitioner does not point to any evidence to support Dr. Gloor’s proclaimed speculative belief that Gloor was accessible on arXiv.org or by any other means at any point in 2010. **Third**, there is no reliable evidence regarding arXiv.org’s policies or business practices at the time Gloor was allegedly submitted that could make up for the deficient direct evidence regarding Gloor’s alleged accessibility on arXiv.org.

PO Resp. 17. We are not persuaded by Patent Owner’s argument.

Exhibit E to the Gloor declaration shows that the Gloor reference was assigned an arXiv identifier as of July 29, 2010. Ex. 1002, Exhibit E. It provides a hyperlink to the abstract and states that the abstract would appear in that day’s mailing. *Id.* In Exhibit F to the Gloor declaration, the arXiv Technical Support states that the submission with the same hyperlink “would have been announced on Thursday, July 29, 2010.” Ex. 1002, Exhibit F.

Patent Owner argues that Exhibit E is an email Dr. Gloor received, and thus, “shows only that Dr. Gloor might have been able to access the

paper.” PO Resp. 18. Exhibit H to the Gloor declaration is a poster presented by Dr. Gloor at the Human Microbiome Research Conference, held between August 31 and September 2, 2010. Ex. 1002 ¶ 18, Exhibits G, H. Dr. Gloor testifies that the poster “describ[es] the findings reported in *Gloor*.” *Id.* ¶ 18. He also testifies that he discussed Exhibit H with other attendees of the conference. *Id.*

Patent Owner contends that “Dr. Gloor’s alleged recollection that he presented *findings* from his paper to attendees of the Human Microbiome Research Conference in August and September 2010 does not demonstrate that the Gloor *reference* was publicly accessible before December 23, 2010.” PO Resp. 19–20. Patent Owner argues that the “mere presentation of slides accompanying an oral presentation at a professional conference is not per se a ‘printed publication’ for the purposes of § 102(b).” *Id.* at 20 (citing Klopfenstein, 380 F.3d at 1349).

As we understand it, however, Petitioner does not present Exhibit H as a printed publication. Nor does it rely on the substantive teachings of Exhibit H. Instead, Petitioner points to the “References section” of Exhibit H, which provides a complete citation to the Gloor reference, to support its position that Gloor was publicly available before December 2010. *See* Reply 28 (citing Ex. 1002 ¶ 18, Exhibits G, H). We find this evidence persuasive as supporting the public availability of Gloor.

Petitioner also presents evidence showing that Gloor was discussed on mothur.org. Ex. 1052. Mothur is a software platform that “provide[s] a flexible and powerful software package for analyzing sequencing data.” Ex. 1053, 1. It includes a discussion forum that “allows users to ask questions that anyone can answer.” *Id.* at 2. Exhibit 1052 shows that, in

October 2010, in a string of discussions of “paired-end reads pipeline (Illumina),” a user wrote that

there is another paper where they developed their own pipeline to analyze paired end[]. It was published in PLoS ONE in august 2010 vol 5 by R Hummelen One of the authors was Gregory Gloor (I believe he developed the code) and he is in the process of publishing a methodology paper on analyzing paired ends.

You can find the paper at [http //arxiv.org/abs/1007.5075](http://arxiv.org/abs/1007.5075).

Ex. 1052, 1. We find this evidence also persuasive to show the public availability of the Gloor reference before December 2010.

The record includes other evidence that supports Petitioner’s assertion that Gloor was publicly available. For example, additional communications with arXiv support Gloor was made public on July 29, 2010. *See* Ex. 1039, 5 (email from “arXiv Technical Support,” stating “[t]he paper was made public on arxiv.org at 8PM Eastern on 7/29/2010”); Ex. 1043 (email from “arXiv Head of Content & User Support,” stating “[f]or the paper <https://doi.org/10.48550/arXiv.1007.5075> it was posted on 30 July 2010 shortly after midnight UTC (8PM ET)”; Ex. 1050 (email from “arXiv Technical Support,” stating “2010-07-30 sometime after midnight or 1AM UTC has been confirmed by our database records”).

Petitioner also presents “[a]rchived arXiv.org webpages from June and September 2010 (i.e., flanking *Gloor*’s July 2010 submission and publication dates)” to show arXiv.org’s routine practice of publishing articles was within 1-2 business days of submission. Reply 26 (citing Ex. 1046 ¶¶ 1–6; Ex. 1049); Exs. 1047, 1048.

Considering the record evidence as a whole, and heeding the Federal Circuit’s instruction “not to overstate the required degree of certainty” when

evaluating public availability, we find Petitioner has shown, by a preponderance of the evidence, that Gloor was publicly accessible before December 2010, and thus, qualifies as prior art. *See CQV Co. v. Merck Pat. GmbH*, 130 F.4th 1344, 1350 (Fed. Cir. 2025).

2. Gloor's Disclosure

Gloor is titled “Microbiome profiling by Illumina sequencing of combinatorial sequence-tagged PCR products.” Ex. 1005, 1. Gloor describes “a low-cost, high-throughput microbiome profiling method that uses combinatorial sequence tags attached to PCR primers that amplify the rRNA V6 region.” *Id.*; *see also id* at 2 (reporting “the methods used to perform microbiome analysis of the V6 region of 272 clinical samples using the Illumina sequencing technology”). Gloor describes sequencing amplified PCR products “using an Illumina paired-end protocol to generate millions of overlapping reads.” *Id.* at 1. According to Gloor, “[c]ombinatorial sequence tagging can be used to examine hundreds of samples with far fewer primers than is required when sequence tags are incorporated at only a single end.” *Id.*

Gloor added primers and sequence tags to nucleic acids prior to sequencing. *Id.* at 3–6. Gloor “var[ied] the length of the sequence tags between 3 and 6 nucleotides long” and “chose sequence tags to ensure all 4 nucleotides were represented in each of the first four positions of the primers using parameters similar to those in the barcrawl program.” *Id.* at 5. Gloor describes developing custom programs “to identify the overlapping segments of the forward and reverse reads” and to extract “overlapping reads with proper primer sequences and correct sequence tags.” *Id.* at 7. Gloor

inspected sequences with “incorrect sequence tags” to determine the sources of the errors. *Id.* at 7–8.

3. Analysis of Anticipation

Petitioner argues that Gloor discloses each limitation of claims 1–6. Pet. 25–34. Patent Owner contends that Gloor does not disclose all limitations of claim 1. PO Resp. 13–15. Patent Owner also asserts that Gloor does not disclose the additional limitation of claim 5. *Id.* at 15–16. We find Petitioner’s argument persuasive.

a. Claim 1

Petitioner argues that Gloor sequenced nucleic acid analytes of interest derived from 272 clinical samples that had been amplified using “combinatorial sequence tags attached to PCR primers.” Pet. 25 (citing Ex. 1005, Abstract). Each nucleic acid analyte of interest was attached to a pair of sequence tags (i.e. the “identifier sequences” used in claim 1). *Id.* (citing Ex. 1003 ¶ 136; Ex. 1005, 7, Fig. 2). According to Petitioner, Gloor’s combinatorial pairs of sequence tags were unique because “nucleic acid analytes of interest from each sample had their own pairs of identifier sequences, and no two samples shared the same pair.” *Id.* (citing Ex. 1003 ¶ 136; Ex. 1005, Abstract, 2, 6, 7); *id.* at 27 (citing Ex. 1003 ¶ 141; Ex. 1005, Abstract, 2, 6).

Petitioner refers to Gloor’s Figure 2, where the sequence tags were attached to the ends of the nucleic acid analyte of interest. *Id.* at 28 (citing Ex. 1003 ¶ 144; Ex. 1005, 2, 5–7, Fig. 2). Petitioner refers to Gloor’s Table 2 to show the disclosure of sequence tags with “four or more nucleotides and are different.” *Id.* at 29 (citing Ex. 1003 ¶¶ 145, 146; Ex. 1005, 6, Table 2).

Petitioner points out that Gloor sequenced the amplified nucleic acid analytes and used sequences with “*two valid sequence tags* and both primer sequences” in “the remainder of the analysis.” *Id.* at 25–26 (citing Ex. 1003 ¶ 138; Ex. 1005, 7); *id.* at 27 (citing Ex. 1003 ¶ 142; Ex. 1005, 7, Fig. 2); *id.* at 30 (citing Ex. 1003 ¶¶ 148, 149; Ex. 1005, 7, Fig. 2). According to Petitioner, Gloor used the pairs of sequence tags to validate the sequences of the nucleic acid analytes. *Id.* at 25–26 (citing Ex. 1003 ¶¶ 137–139; Ex. 1005, 6–8, 22–23); *id.* at 31–32 (citing Ex. 1003 ¶¶ 150, 151; Ex. 1005, 1, 7–8, 22–23).

Patent Owner argues that Gloor does not disclose “two or more identifier sequences that are uniquely associated with the nucleic acid analyte of interest.” PO Resp. 13–15 (citing Ex. 1001, 2:5–8, 22:2–6; Ex. 1003 ¶ 26; Ex. 2001 ¶¶ 83–85). Patent Owner’s argument relies entirely on its proposed claimed construction. *Id.* at 14 (arguing that “any identifier sequence can be used with only one template nucleic acid of interest” and that “[o]nce an identifier sequence has been used with one nucleic acid analyte of interest, it cannot be used with another—otherwise it would no longer be ‘uniquely’ associated with the nucleic acid of interest”).

As explained above, we reject Patent Owner’s proposed claimed construction. *See supra* Section II.C. Instead, we construe “two or more identifier sequences that are uniquely associated with the nucleic acid analyte of interest” means two or more sequences together are uniquely associated with the nucleic acid analyte of interest, and does not require each identifier sequence be uniquely associated with the nucleic acid analyte of interest. *Id.*

Under our construction, Gloor discloses the disputed limitation. Gloor “used paired-end sequencing in combination with unique sequence tags.” Ex. 1005, 2 (stating it was able to “use a small number of primers to uniquely tag a large number of samples”). Thus, we agree with Petitioner that in Gloor, “the nucleic acid analytes from each sample contained a pair of two identifier sequences that was not shared by any other sample, i.e., the pair was ‘unique’ to that sample.” Pet. 27.

Patent Owner does not dispute that Gloor discloses the remaining limitations of claim 1. *See* PO Resp. 13–15. We find Petitioner’s mapping of the remaining limitations of claim 1 to Gloor persuasive and adopt it as our own. Thus, we determine Gloor anticipates claim 1.

b. Claim 5

Claim 5 recites “t[he] barcode sequences are designed to have no homopolymer regions.”

Pointing to Gloor’s Table 2, Petitioner asserts that the barcode sequences shown there include no consecutively repeated nucleotides, and thus, have no homopolymer regions. Pet. 33 (citing Ex. 1003 ¶¶ 158, 159; Ex. 1005, 6 (Table 2)).

Patent Owner contends that “Gloor’s incidental lack of homopolymer regions in its chosen sequence tags does not show any teaching that ‘t[he] barcode sequences are *designed* to have no homopolymer regions.’” PO Resp. 16 (citing Ex. Ex. 2001 ¶ 88); *see also id.* at 15 (“Gloor’s discussion of its sequence tag design contains *no discussion* of homopolymer regions.”).

We are not persuaded by Patent Owner’s argument. Claim 5’s additional limitation is a negative limitation. “[A] reference need not state a

feature’s absence in order to disclose a negative limitation.” *AC Techs. S.A. v. Amazon.com, Inc.*, 912 F.3d 1358, 1367 (Fed. Cir. 2019); *see also Sud-Chemie, Inc. v. Multisorb Techs., Inc.*, 554 F.3d 1001, 1004–05 (Fed. Cir. 2009) (affirming finding that reference disclosed “uncoated” film where it did not describe the film as coated and did not suggest the necessity of coatings).

We also are not persuaded by Patent Owner’s emphasis on the term “designed.” The disclosed barcode sequences with no homopolymer regions do not exist in nature, and thus, are designed.

In sum, we find Petitioner has shown Gloor discloses the additional limitation of claim 5, and thus, anticipates claim 5.

c. Other Claims

Petitioner argues that Gloor discloses each limitation of claims 2–4 and 6. Pet. 32–34. Patent Owner does not dispute these assertions. *See generally* PO Resp. We find Petitioner’s mapping of the limitations to Gloor persuasive, and adopt it as our own. Thus, we determine Gloor anticipates claims 2–4 and 6.

E. Alleged Anticipation by Parameswaran

Petitioner asserts that claims 1–6 are anticipated by Parameswaran. Pet. 34–43. After reviewing the entire record developed at trial, and as explained below, we determine Petitioner has not shown by a preponderance of the evidence that Parameswaran anticipates claims 1–6.

1. Parameswaran’s Disclosure

Parameswaran is titled “A pyrosequencing-tailored nucleotide barcode design unveils opportunities for large-scale sample multiplexing.” Ex. 1004, 1. Parameswaran uses 454/Roche Genome Sequencers for

sequencing. *Id.* The technology of these pyrosequencers is based on the detection of pyrophosphates released during DNA synthesis. *Id.*

Parameswaran describes “a high-information-content barcoding approach in which each sample is associated with two uniquely designed, 10-nucleotide barcodes.” *Id.* at 2. According to Parameswaran, “[t]he presence of these assigned barcodes allow[s] for independent samples to be pooled together for sequencing, with subsequent bioinformatic segregation of the pyrosequencer output.” *Id.* In addition to barcodes, Parameswaran incorporated forward and reverse cloning linkers into nucleic acids. *Id.* at 2–3.

Figure 1 of Parameswaran, reproduced below, depicts the design of forward and reverse primers. *Id.* at 3.

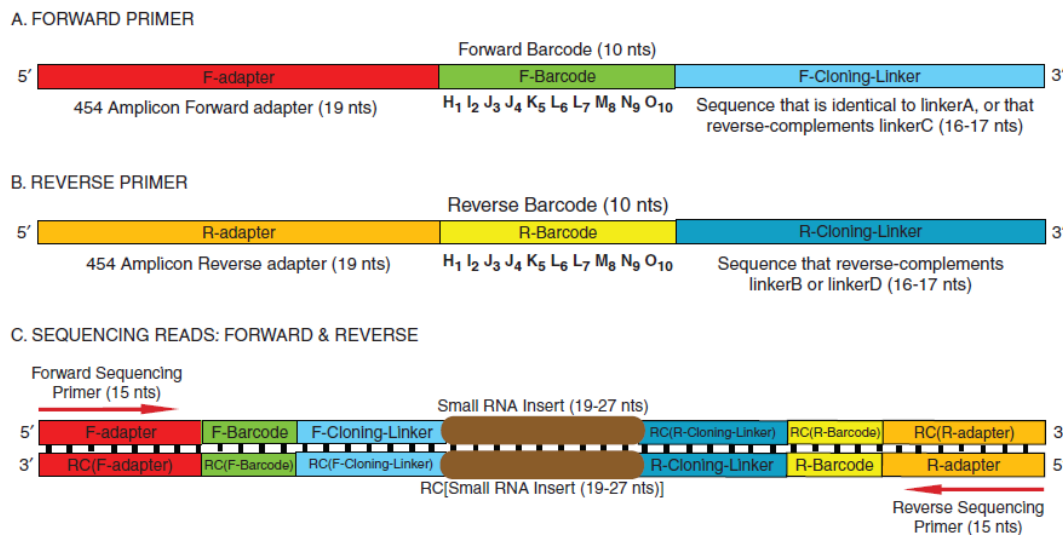


Figure 1. Design of forward and reverse primers. The synthesized primers are 45–46 nt long. (A and B) The template for the primers is: 454-Adapter:: Barcode:: Linker-Primer. Individual specifications for forward and reverse barcodes are also indicated. (C) Diagrammatic representation of the GS20 forward and reverse sequencing reads. ‘RC’ stands for Reverse Complement. Teeth denote base pairing.

Id. at 3. Parameswaran’s Figure 1, above, shows that a forward primer comprises the Roche-specific forward adapter (“F-adapter”), the forward barcode (“F-Barcode”), and the forward cloning linker (“F-Cloning-Linker”). *Id.* It also shows that a reverse primer comprises the

Roche-specific reverse adapter (“R-adapter”), the reverse barcode (“R-Barcode”), and the reverse cloning linker (“R-Cloning-Linker”). *Id.*

Parameswaran describes expanding the 10-nucleotide barcodes to “barcode motifs,” which included three additional nucleotides from linkers adjoining the barcodes, to maximize the uniqueness of dual-indexing and facilitate definitive sample assignments. *Id.* at 7. Parameswaran’s study aimed to determine “how much of the (fixed) linker sequence will be required in addition to the (variable) barcode for definitive sample assignment.” *Id.* at 7–8. Ultimately, “a search motif of 13 nt was determined to be ideal for our studies in minimizing false-discovery rate.” *Id.*

Parameswaran reports that using 13-nt barcode motifs, approximately 94% of sequences were identified to have “perfect forward and reverse barcode motifs.” *Id.* at 7. Parameswaran subdivided the sequences having a perfect match to only one of the expected barcode motifs into three classes. *Id.* One of the three classes includes those having “perfect identity to an unexpected motif, resulting in an unassigned 5’–3’ motif combination (perfect, unexpected occurrence).” *Id.* Parameswaran reports that “[t]he false-discovery rate, defined by the percentage of ‘perfect and unexpected’ occurrences, was observed to decrease to <0.005%.” *Id.* at 8.

Parameswaran speculates that its pyrosequencer-tailored barcodes may also be compatible with other sequencing platforms. *Id.* at 2, 8. Parameswaran specifies that Illumina’s platform is one of three major high-throughput sequencing platforms. *Id.* at 1.

2. Analysis of Anticipation

Petitioner argues that Parameswaran discloses each limitation of claims 1–6. Pet. 34–34. Patent Owner contends that Parameswaran does not

disclose all limitations of claim 1. PO Resp. 23–25. As explained below, although we agree with Petitioner that Parameswaran discloses steps i) and ii) of claim 1, we are persuaded by Patent Owner’s argument that Parameswaran does not disclose step iii), the validating step.

Petitioner argues that Parameswaran used Roche 454 multiplex sequencing to sequence “DNA from two independent sets of 25 barcoded libraries” prepared from multiple samples. Pet. 34 (citing Ex. 1003 ¶ 163; Ex. 1004, 7). Two “uniquely designed, 10-nucleotide barcodes” and nucleotide linkers were attached to the ends of the nucleic acids from each sample. *Id.* at 35–39 (citing Ex. 1003 ¶¶ 163, 168–174; Ex. 1004, 2, 3, 7, 8, 14–17).

Petitioner points out that, after sequencing, Parameswaran evaluated the sequences using barcode motifs. *Id.* at 35, 39–40 (citing Ex. 1003 ¶¶ 163, 175, 176; Ex. 1004, 3, 7–8). In Parameswaran, about 94% of the sequences had perfect forward and reverse barcode motifs, while about 5–6% had either only a single identifier sequence or a pair of mismatched identifier sequences. *Id.* at 35, 40 (citing Ex. 1003 ¶¶ 164, 165, 176, 177; Ex. 1004, 7–8).

Petitioner emphasizes that Parameswaran uses terms such as “false-discov[ies]” and “misassignment” to describe the erroneous sequences. *Id.* at 35–36, 40 (citing Ex. 1003 ¶¶ 165, 166, 177; Ex. 1004, 7–8). Parameswaran explains that various errors may have produced the misassignments. *Id.* at 36, 40 (citing Ex. 1003 ¶¶ 166, 177; Ex. 1004, 7–8).

According to Petitioner,

Parameswaran characterizes sequences with mismatched barcodes as “false-discover[ies]” arising due to various errors that result in “misassignment” of sequences to incorrect

samples. EX1004, 7-8; EX1003, ¶¶178-179. And *Parameswaran* expressly links sequencing accuracy with the presence of both barcode motifs, stating that “any sequence with **both 5’ and 3’ barcode motifs** will be placed with reasonable accuracy into the correct bin.” EX1004, 8; EX1003, ¶179. Thus, *Parameswaran* discloses that lowering the misassignment rate allows sequences to be assigned to correct samples more accurately. EX1004, 8.; EX1003, ¶179.

Parameswaran therefore discloses “validating the sequence of the nucleic acid analyte of interest by analyzing both identifiers and excluding the sequences of those nucleic acid analytes of interest containing only one identifier or an incorrect pair of identifiers from sequence analysis of the multiplex sequencing reaction.” EX1003, ¶180.

Id. at 41. We disagree with Petitioner’s conclusion.

Parameswaran states that “[t]he low false-discovery rate for perfect matches strengthens the argument that any sequence with both 5’ and 3’ barcode motifs will be placed with reasonable accuracy into the correct bin.” Ex. 1004, 8. But, because *Parameswaran* achieved a “remarkably low misassignment rate,” it did not exclude, but included any sequence with a matched barcode. *Id.*

Indeed, *Parameswaran* explains that “[f]or our purposes, all sequences with a match to one of the two expected sequence motifs were considered as true members of the corresponding dataset.” *Id.* In its Reply, Petitioner contends that *Parameswaran* refers to sequences with “**expected** sequence motifs,” and not “perfect, **unexpected** occurrences.” Reply 6 (citing Ex. 1004, 7, 8; Ex. 1054 ¶¶ 26, 27; Ex. 1055, 52:15–53:5). We are not persuaded by this argument.

Parameswaran included “all sequences with a match to **one of the two expected** sequence motifs.” Ex. 1004, 8 (emphasis added). In other words, sequences with one expected sequence motif and another unexpected

sequence motif were included. Indeed, Parameswaran explicitly states that only “sequences [that] had no perfect matches to any of the sequence motifs (forward or reverse) . . . were discarded.” *Id.*

Pointing to Parameswaran’s supplementary table, Petitioner also argues that sequences with mismatched barcode motifs were greyed out, “signifying that they were excluded.” Reply 6–7 (citing Ex. 1004, 23, 29; Ex. 1054 ¶¶ 28–34). We are not persuaded by this argument either.

Dr. Spellman testified that, in the supplementary table, the discarded sequences correspond to the four blank cells in the bottom right of the table, not to the greyed-out cells. Ex. 2015, 102:3–104:19. Neither Dr. Spellman nor Petitioner explains why the grayed boxes they highlight were not simply left blank if those sequences were actually discarded, like the four boxes in the data table corresponding to sequences Parameswaran actually states were discarded.

In sum, despite labelling sequences with “mismatched 5’ and 3’ motifs (i.e. sequences that do not have corresponding 5’ and 3’ motifs)” as “false-discover[ies]” with “misassignment[s],” Parameswaran included, and did not exclude, the sequences with perfect, but unexpected barcode motifs. Ex. 1004, 8. Thus, Parameswaran does not disclose the validating step of claim 1. As a result, Petitioner has not shown by a preponderance of the evidence that Parameswaran anticipates claim 1.

Each of claims 2–6 depends from claim 1. Thus, Petitioner has not shown by a preponderance of the evidence that Parameswaran anticipates claims 2–6.⁷

F. Alleged Obviousness over Parameswaran and Gloor

Petitioner asserts that claims 1–6 would have been obvious over the combination of Parameswaran and Gloor. Pet. 43–56. Petitioner argues that each reference teaches every limitation of the challenged claims. *Id.* at 43–48, 54–56 (citing Ex. 1003 ¶¶ 191, 193–201). Petitioner also asserts that “[a] POA would have been motivated to combine these references, with a reasonable expectation of success, because both use dual barcoding strategies for multiplex sequencing; both are directed to reducing sequencing error; and both emphasize the compatibility of dual indexing for Roche 454 and Illumina sequencing to reduce sequencing error.” *Id.* at 43–44 (citing Ex. 1003 ¶ 192); *id.* at 48–53 (citing Ex. 1003 ¶¶ 202–215).

Patent Owner argues that Petitioner fails to show a motivation to combine and fails to show a reasonable expectation of success. PO Resp. 27–46 (citing Ex. 2001 ¶¶ 58–60, 65–75, 101–141). Patent Owner also contends that the combined teachings of Parameswaran and Gloor do not disclose the additional limitation of claim 5. *Id.* at 46–47 (citing Ex. 2001 ¶¶ 86–88, 92–96, 142).

As explained above, we determine Gloor discloses each limitation of claims 1–6, and thus, anticipate these claims. *See supra* Section II.D.3.

⁷ Patent Owner also asserts that Parameswaran does not disclose the additional limitations of claims 5 and 6. PO Resp. 25–27. Because we determine that Petitioner has not shown Parameswaran discloses all the limitations of claim 1, we do not need to address these arguments.

Because “anticipation is the epitome of obviousness,” we conclude that the combination of Parameswaran and Gloor renders claims 1–6 obvious. *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1373 (Fed. Cir. 2019) (affirming the Board’s obviousness determination based on an anticipatory prior art alone and stating because the Board did not rely on the other reference “for the disclosure of a particular element or teaching, the Board had no obligation to find a motivation to combine”).

G. Alleged Obviousness over Parameswaran and Binladen

Petitioner asserts that claims 1–6 would have been obvious over the combination of Parameswaran and Binladen. Pet. 56–66. After reviewing the entire record developed at trial, and as explained below, we determine Petitioner has shown, by a preponderance of the evidence, that the combined teaching of Parameswaran and Binladen renders claims 1–4 obvious. We determine Petitioner has not shown, by a preponderance of the evidence, that the combined teaching of Parameswaran and Binladen renders claims 5 or 6 obvious.

1. Binladen’s Disclosure

Binladen used “conventional PCR with 5’-nucleotide tagged primers to generate homologous DNA amplification products from multiple specimens, followed by sequencing through the high-throughput Genome Sequence 20TM DNA Sequencing System (GS20, Roche/454 Life Sciences).” Ex. 1029, 1. “Each DNA sequence is subsequently traced back to its individual source through 5’ tag-analysis.” *Id.* Binladen states that its approach “enables the assignment of virtually all the generated DNA sequences to the correct source once sequencing anomalies are accounted for (miss-assignment rate<0.4%).” *Id.*

In Binladen, “initial PCR primers are 5’-tagged with short nucleotide sequences (tags) in such a way that a unique tagged primer combination can be applied to each specific DNA template source.” *Id.* at 2. Binladen used both dinucleotide tags and tetranucleotide tags. *Id.* at 3, Table 1.

Binladen used DNA from thirteen species as PCR template. *Id.* at 3. Binladen explains that “under the [then-]current status of the sequencing technology, GS20 sequencing reads are limited to approximately 120 bases, thus in this experiment the full sequence (133–141 bp including primer, species dependent) was not returned.” *Id.* According to Binladen, “[t]he implication of this is that approximately 50% of each PCR product will be sequenced from the orientation of the forward primer, and 50% from the orientation of the reverse primer. Hence, this made it necessary to label both the forward and reverse end of each PCR product.” *Id.*

Binladen states that because “the correct association of tags and sequences is crucial to the approach,” it “discarded all sequence reads without an exact match to any of the primers used in the studies (Primer Mismatched Sequences).” *Id.* at 5. “Subsequently, the identity of the remaining sequences were globally aligned to the thirteen reference sequences (Sanger-sequencing generated) using direct and reverse complementation.” *Id.* Binladen reports that 89% of the remaining sequences “diverged by no more than 1bp from one of the reference sequences, and could thus be assigned to one of the taxa analysed in the study.” *Id.* at 6.

2. Analysis

Petitioner argues that the combination of Parameswaran and Binladen teaches each limitation of claims 1–6. Pet. 57–62, 64–66. Petitioner also asserts that “[a] POSA would have been motivated, with a reasonable

expectation of success, to combine *Parameswaran* and *Binladen* to improve multiplex sequencing accuracy.” *Id.* at 62.

Patent Owner contends that the asserted combination does not teach all limitations of claims 1, 5, and 6. PO Resp. 47–52. Patent Owner also asserts that Petitioner fails to show a motivation to combine *Parameswaran* and *Binladen*. *Id.* at 15–16.

For claims 5 and 6, we find Patent Owner’s argument persuasive. For claims 1–4, we find Petitioner’s argument persuasive.

a. Claim 1

As explained above, we agree with Petitioner that *Parameswaran* discloses steps i) and ii) of claim 1. *See supra* Section II.E.2; *see also* Pet. 34–39 (Petitioner’s detailed mapping of claim 1’s steps i) and ii) to *Parameswaran*’s disclosures).

Also as explained above, we reject Petitioner’s anticipation challenge based on *Parameswaran*, because *Parameswaran* does not disclose excluding mismatched sequences, as required by the validating step of claim 1. *See supra* Section II.E.2 (citing Ex. 1004, 8). An obviousness determination, however, involves a different inquiry: we ask whether Petitioner has shown a POSA would have had a reason to modify *Parameswaran* to exclude relevant sequences. As explained below, on this record, we find Petitioner has met its burden in this showing. *See* Pet. 40–41, 46–47, 61.

As Petitioner correctly points out, even though *Parameswaran* included “all sequences with a match to one of the two expected sequence motifs,” it “characterizes sequences with mismatched barcodes as ‘false-discover[ies]’ arising due to various errors that result in ‘misassignment’ of sequences to incorrect samples.” Pet. 40–41 (citing

Ex. 1003 ¶¶ 178, 179; Ex. 1004, 7–8). We are persuaded that a POSA would have recognized that “mismatched sequences are erroneous and that reducing the rate of such ‘false discovery’ would decrease the rate at which sequences are assigned to incorrect samples.” *See* Ex. 1003 ¶ 179 (citing Ex. 1004, 7–8) (“*Parameswaran* calls out the low incidence of such errors when requiring a match to the 13-base barcode motif as an advantage of its unique dual-barcoding method.”). *Parameswaran* also states that “[t]he low false-discovery rate for perfect matches strengthens the argument that any sequence with both 5’ and 3’ barcode motifs will be placed with reasonable accuracy into the correct bin.” Ex. 1004, 8. Thus, Petitioner has shown that a POSA would have had a reason to modify *Parameswaran* to exclude relevant mismatched sequences.

On this record, Petitioner has shown that in addition to disclosing steps i) and ii) of claim 1, *Parameswaran* also suggests excluding relevant mismatched sequences, as required by step iii) of claim 1. We are also persuaded that a POSA following the modified teachings of *Parameswaran* would have had a reasonable expectation of success to validate the sequence of nucleic acid analyte of interest in a multiplex sequencing reaction.

Petitioner does not rely on *Binladen* for any particular element or teaching missing from *Parameswaran*. *See* Pet. 57–61. Therefore, we do not address Patent Owner’s argument challenging Petitioner’s showing of the motivation to combine *Parameswaran* and *Binladen*. *See Realtime Data*, 912 F.3d at 1373.

In sum, we find Petitioner has shown, by a preponderance of the evidence, that claim 1 would have been obvious over the combination of *Parameswaran* and *Binladen*.

b. Claims 2–4

Petitioner argues that each of Parameswaran and Binladen discloses the additional limitations of claims 2–4. Pet. 64–65. Patent Owner does not dispute these assertions. *See generally* PO Resp. We find Petitioner’s mapping of the limitations to the asserted prior art persuasive and adopt it as our own. Thus, we find Petitioner has shown, by a preponderance of the evidence, that claims 2–4 would have been obvious over the combination of Parameswaran and Binladen.

c. Claim 5

Claim 5 recites “t[he] barcode sequences are designed to have no homopolymer regions.” As explained above, the term “homopolymer region” means “two or more of the same base in a row,” as the ’200 patent specification expressly defines. *See supra* Section II.C (citing Ex. 1001, 9:6–10).

Petitioner argues that Parameswaran teaches designing the barcode “to avoid homopolymeric runs.” Pet. 43 (citing Ex. 1004, 4). Parameswaran, however, only states that its barcodes have “no homopolymer *beyond* 2 nt.” Ex. 1004, 7 (emphasis added). In other words, Parameswaran’s barcodes may have two of the same base in a row, which is a “homopolymer region” under the ’200 patent’s express definition. And as Patent Owner points out, every one of Parameswaran’s barcodes contains at least one such homopolymer region. PO Resp. 25 (citing Ex. 1004, Table 2).

Parameswaran states that “[w]ith the barcode design used in this work (i.e. no homopolymer beyond 2 nt), we expected to minimize errors related to pyrosequencing. Indeed, the experimentally observed spectrum of errors did not match the types of homopolymer-related errors most frequently

observed for pyrosequencing.” Ex. 1004, 7. Petitioner has not sufficiently explained why a POSA, reading Parameswaran, would have had a reason to avoid homopolymer region with two of the same base in a row.

Petitioner asserts that Binladin’s tetranucleotide tags do not have homopolymer region. Pet. 65 (citing Ex. 1029, 3, 7, Tables 1, 3). Although that statement is correct, we are not persuaded that, a POSA would have adopted Binladin’s tetranucleotide tags.

As Petitioner argues,

Binladen used 2-nucleotide and 4-nucleotide long sequence identifiers, and observed that the longer identifiers lowered the misassignment rate. EX1029, 7; EX1003, ¶244. *Parameswaran* made this explicit, characterizing Binladen as “valuable” but explaining that some applications “call for a more definitive assignment of samples and/or for enhancing multiplexing capabilities,” and that this could be achieved through the use of “10-base barcodes that differ from each other at multiple positions (by at least 4 nt).” EX1004, 2; EX1003, ¶244.

Pet. 63; *see also* Reply 24–25 (arguing that the fact that Binladen’s barcodes are short “**strengthens** the motivation to combine,” because “[t]o improve multiplex sequencing accuracy, a POSA would have been motivated to use *Parameswaran*’s unique dual-indexing method, including its longer barcodes”). Petitioner has not sufficiently explained why, a POSA, understanding the advantage of *Parameswaran*’s “10-base barcodes,” would have adopted Binladen’s tetranucleotide tags.

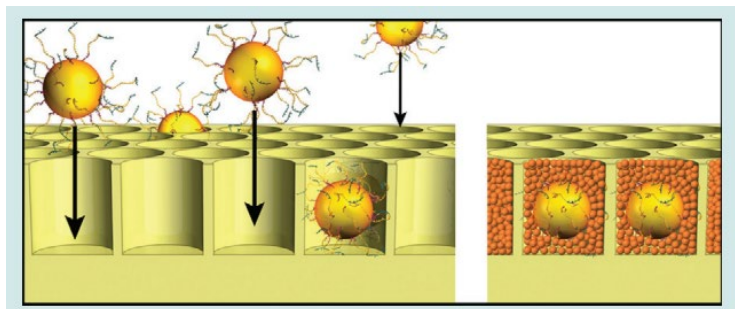
In sum, on this record, we find Petitioner has not shown sufficiently that the asserted references teach the additional limitation of claim 5. Thus, Petitioner has not shown by a preponderance of the evidence that claim 5 would have been obvious over the combination of *Parameswaran* and *Binladen*.

d. Claim 6

Claim 6 recites “the nucleic acid analyte of interest is on the surface of a flow cell.”

Petitioner acknowledges that Parameswaran and Binladen both use Roche 454 sequencing. Pet. 66. According to Petitioner, Roche 454 sequencing involves placing “bead-bound nucleic acid analytes in wells of a flow cell.” *Id.* (citing Ex. 1028, 6–8). Petitioner argues that “[c]laim 6 does not require attaching the analyte to the surface of a flow cell—it simply requires the analyte to be “on the surface of a flow cell.” *Id.* at 65–66 (citing Ex. 1003 ¶ 251). Thus, Petitioner contends that both references teach the additional limitation of claim 6. *Id.* We are not persuaded by this argument.

In Roche 454 sequencing, the analyte of interest is bound to a bead and deposited into a flow cell, as shown below.



The figure above shows deposition of DNA beads into the wells of a plate. Ex. 1028, 8; *see also* Ex. 1010, 4 (“The beads are incubated with a polymerase and single-strand binding proteins and, together with smaller beads carrying the ATP sulfurylases and luciferases, gravitationally deposited in the wells.”).

We find persuasive Patent Owner’s argument that “[d]epositing analyte-bound beads into wells was used in Roche 454 sequencing because it permits the beads to be free floating and surrounded by reagents to provide

light to read the analyte of interest.” *See* PO Resp. 27 (citing Ex. 2001 ¶¶ 99, 100). Thus, we agree with Patent Owner that the nucleic acid analytes bound to the beads are suspended in the well, and not “on the surface of a flow cell.” *Id.* at 26–27.

In sum, on this record, we find Petitioner has not shown sufficiently that the asserted references teach the additional limitation of claim 6. Thus, Petitioner has not shown by a preponderance of the evidence that claim 6 would have been obvious over the combination of Parameswaran and Binladen.

H. Alleged Obviousness over Parameswaran, Binladen, and Bentley

Petitioner asserts that claim 6 would have been obvious over the combination of Parameswaran, Binladen, and Bentley. Pet. 66–67. After reviewing the entire record developed at trial, and as explained below, we determine Petitioner has shown by a preponderance of the evidence that the challenged claims would have been obvious over the asserted prior art.

1. Bentley

Bentley provides a detailed description of Illumina paired-end sequencing. Ex. 1006. It describes a “massively parallel synthetic sequencing approach” that involved shearing DNA strands into double-stranded DNA fragments for sequencing and ligating (joining) Illumina-specific adapters to both ends. *Id.* at 1–2, 13–14. Bentley “generated high-density single-molecule arrays of genomic DNA fragments attached to the surface of the reaction chamber (the flow cell) and used isothermal ‘bridging’ amplification to form DNA ‘clusters’ from each fragment.” *Id.* at 1. It describes that these DNA fragments were then denatured into single strands so that Illumina-specific adapters on the ends

of the templates annealed to complementary oligonucleotides on the surface of the flow cell. *Id.* at 3.

2. Analysis

Petitioner argues, and Patent Owner does not dispute, that Bentley teaches Illumina sequencing of template nucleic acids after bridge amplification on the surface of the flow cell. Pet. 66 (citing Ex. 1006, 1–2).

Petitioner contends that “to increase accuracy and decrease cost, a POSA would have been motivated to use the dual index strategies of *Parameswaran* and *Binladen* with Illumina sequencing as disclosed in *Bentley*.” *Id.* at 66–67 (citing Ex. 1003 ¶ 254). Patent Owner disagrees. PO Resp. 27–46, 52–54. We find Petitioner’s argument more persuasive.

As Petitioner correctly points out, *Parameswaran* states that its barcoding approach “may also be compatible with other sequencing platforms,” and specifically identifies Illumina sequencers as one of the three “major high-throughput sequencing platforms.” Pet. 67 (citing Ex. 1003 ¶ 255; Ex. 1004, 1, 2, 8).

Patent Owner contends that a POSA would not have had a reason to use *Parameswaran*’s unique dual indexes in Bentley’s Illumina sequencers because “Illumina sequencers did not have the same needs of pyrosequencing that guided the design of *Parameswaran*.” PO Resp. 28 (citing Ex. 2001 ¶¶ 102–105). Patent Owner also asserts that *Parameswaran*’s approach does not apply to Illumina sequencers because *Parameswaran*’s barcodes are “tailored” for pyrosequencing. *Id.* at 29–30 (citing Ex. 1004, 1; Ex. 2001 ¶¶ 104–105). Patent Owner further argues that “a POSA at the time of the invention would not have understood chimeric sequences giving rise to such errors, such as the cross-over error discussed

throughout the '200 Patent, to be a problem in pyrosequencers like Parameswaran's or Illumina sequencers" like Bentley's. *Id.* at 36. We are not persuaded by Patent Owner's arguments.

First, the challenged claims do not recite "chimeric sequences" or "cross-over error." Instead, they recite validating the sequence of the nucleic acid analyte of interest by excluding sequences "containing only one identifier or an incorrect pair of identifiers." Thus, whether chimeric sequences giving rise to cross-over error was known or not is irrelevant to our analysis here.

Second, Parameswaran recognizes that "[o]ptimally, however the precise structure of the barcodes might need to be revised to reflect sequencing error profiles of the various systems." Ex. 1004, 8. Petitioner presents evidence to show that "in December 2010, barcodes posed no barrier for switching from one NGS platform to another; a POSA understood how to design barcodes suitable for Illumina sequencers." Reply 17 (citing Ex. 1054 ¶¶ 101, 102; Ex. 1055, 104:13–105:11). Indeed, samples can be sequenced using the same set of barcodes on both Roche 454 and Illumina platforms. *Id.* at 18 (citing Ex. 1038, Table 1). Thus, we are not persuaded by Patent Owner's argument that Parameswaran's approach applies only to pyrosequencing because its barcodes are "tailored" for pyrosequencing. *See* PO Resp. 29–30.

Third, Petitioner has presented evidence to show that increasing accuracy and decreasing cost is a universal goal in multiplex sequencing,

regardless of the platform. For example, Fuller⁸ recognized the challenge to reduce the cost of ensemble-based sequencing-by-synthesis (“SBS”).

Ex. 1059, 1. Ensemble-based sequencing involves “the sequencing of multiple i[]dential copies of a DNA molecule, typically amplified together on isolated *surfaces or beads*.” *Id.* (emphasis added). In other words, ensemble-based sequencing includes sequencing on Illumina sequencers. *See id.*, Figure 1 (citing Bentley). Fuller cites Parameswaran as teaching “an effective technology to judiciously add sequence tags to the fragments of several different samples.” According to Fuller, “[t]his reduces the inefficiencies of ensemble-based SBS architectures.” *Id.* at 3. Fuller does not limit application of Parameswaran’s method to Roche 454 pyrosequencing only.

Other prior art references similarly suggest that Parameswaran’s method is not limited to Roche 454 pyrosequencing only. *See, e.g.*, Ex. 1062, 11 (citing Parameswaran’s strategy for “allow[ing] the combination of multiple whole community assays in a single run, thereby reducing the per sample cost” in next-generation sequencing, which includes both Roche 454 and Illumina sequencer); Ex. 1068, 7. Thus, we agree with Petitioner that a POSA would have had a reason to use Parameswaran’s dual-index strategies in Bentley’s Illumina sequencing

As a result, Petitioner has shown by a preponderance of the evidence that claim 6 would have been obvious over the combination of Parameswaran, Binladen, and Bentley.

⁸ Fuller, et al., *The Challenges of Sequencing by Synthesis*, 27 NATURE BIOTECHNOLOGY (11) 1013-1023 (2009) (Ex. 1059, “Fuller”).

III. CONCLUSION

Petitioner has met its burden to show that (1) Gloor anticipates claims 1–6; (2) the combination of Parameswaran and Gloor renders claims 1–6 obvious; (3) the combination of Parameswaran and Binladen renders claims 1–4 obvious; and (4) the combination of Parameswaran, Binladen, and Bentley renders claim 6 obvious.

Petitioner has not shown, by a preponderance of the evidence, that (1) Parameswaran anticipates claims 1–6; or (2) the combination of Parameswaran and Binladen renders claims 5 and 6 obvious.

In summary:

Claims	35 U.S.C. §	References	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–6	102	Gloor	1–6	
1–6	102	Parameswaran		1–6
1–6	103	Parameswaran, Gloor	1–6	
1–6	103	Parameswaran, Binladen	1–4	5, 6
6	103	Parameswaran, Binladen, Bentley	6	
Overall Outcome			1–6	

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has proven, by a preponderance of the evidence, that claims 1–6 of the '200 patent are unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

FOR PETITIONER:

William Raich
Charles Collins-Chase
FINNEGAN, HENDERSON, FARABOW,
GARRETT & DUNNER, LLP
william.raich@finnegan.com
charles.collins-chase@finnegan.com

FOR PATENT OWNER:

Brian Matty
Bridget Belcher
DESMARAIS LLP
bmatty@desmaraisllp.com
bbelcher@desmaraisllp.com